SIMPLIFYING EXPRESSIONS

BASIC ALGEBRA

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GRADING SYSTEM

FINAL GRADE

30%

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SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Algebraic Expressions

Properties of Real Numbers

Exponent Rules

Radical Rules

ALGEBRAIC EXPRESSIONS

An expression made up of **letters** and/or **numbers** with sign(s)

$$\mathbf{2x + 3y - 1}$$

VARIABLE

Represents a number with **varying** values

$$2\boldsymbol{x} + 3\boldsymbol{y} - 1$$

CONSTANT

Represents a number with **fixed** value

$$2x + 3y - 1$$

NUMERICAL COEFFICIENT

The constant number that **multiplies** a variable

$$\mathbf{2}x + \mathbf{3}y - \mathbf{1}$$

A combination of variable(s) and constant(s) which is separated by **addition** or **subtraction**

$$\boxed{2x} + \boxed{3y} - \boxed{1}$$

PROPERTIES OF REAL NUMBERS

COMMUTATIVE LAW

Addition	Multiplication
$a + b = b + a$	$ab = ba$
$2 + 3 = 3 + 2$	$(2)(3) = (3)(2)$

ASSOCIATIVE LAW

Addition

$$a + (b + c) = (a + b) + c$$

$$1 + (2 + 3) = (1 + 2) + 3$$

Multiplication

$$a(bc) = (ab)c$$

$$(2)(3 \cdot 4) = (2 \cdot 3)(4)$$

DISTRIBUTIVE LAW OF MULTIPLICATION

$$a(b+c) = ab + ac$$

$$2(3+2) = (2 \cdot 3) + (2 \cdot 2)$$

EXPONENT RULES

PRODUCT RULE

$$2^2 \cdot 2^3 \quad = 2^{2+3}$$

$$x^n \cdot x^m = x^{n+m}$$

In multiplying two expressions with the same base, **add** the exponents

QUOTIENT RULE

$$\frac{2^4}{2^2} = 2^{4-2}$$

$$\frac{x^n}{x^m} = x^{n-m}$$

In dividing two expressions with the same base, **subtract** the exponents

POWER RULE

$$(2^2)^3 = 2^{2\cdot 3}$$

$$(x^n)^m = x^{n\cdot m}$$

In raising a power to a power, **multiply** the exponents

POWER OF ZERO

$$\frac{2^3}{2^3} = 1$$

$$x^0 = 1$$

Any number raised to zero is 1,

except $\mathbf{0^0}$ which is **undefined**

NEGATIVE EXPONENTS

$$2^{-2} = \frac{1}{2^2}$$

$$x^{-n} = \frac{1}{x^n}$$

Any number raised to a negative power is equal to the **reciprocal** of that number raised to the positive power

FRACTIONAL EXPONENTS

$$4^{\frac{1}{2}} = \sqrt[2]{4}$$

$$x^{\frac{1}{n}} = \sqrt[n]{x}$$

Any number raised to the power of $\frac{1}{n}$ is equal to the ***n-th* root** of that number

Distribute an **EXPONENT** over a **PRODUCT**

$$(2 \cdot 3)^2 = 2^2 \cdot 3^2$$

$$(x \cdot y)^n = x^n \cdot y^n$$

The power of a product is equal to the product of the powers

$$(x + y)^n \neq x^n + y^n$$

$$(x - y)^n \neq x^n - y^n$$

Distribute an **EXPONENT** over a **PRODUCT**

$$(2 \cdot 3)^2 = 2^2 \cdot 3^2$$

$$(x \cdot y)^n = x^n \cdot y^n$$

The power of a product is equal to the product of the powers

$$(2 + 3)^2 \neq 2^2 + 3^2$$

Distribute an **EXPONENT** over a **QUOTIENT**

$$\left(\frac{1}{2}\right)^2 = \frac{1^2}{2^2}$$

$$\left(\frac{x}{y}\right)^n = \frac{x^n}{y^n}$$

The power of a quotient is equal to the quotient of the powers

Simplify the following expression:

$$\frac{3x^{-2}}{x^4}$$

Simplify the following expression:

$$\frac{4y^3}{y^{-5}}$$

Simplify the following expression:

$$\frac{y^3 z^5}{7z^{-2}y^7}$$

RADICAL RULES

RADICAL OF A PRODUCT

$$\sqrt{4 \cdot 9} \quad = \sqrt{4} \cdot \sqrt{9}$$

$$\sqrt[n]{a \cdot b} = \sqrt[n]{a} \cdot \sqrt[n]{b}$$

The radical of the product is the product of the radicals

$$\sqrt[n]{a + b} \neq \sqrt[n]{a} + \sqrt[n]{b}$$

$$\sqrt[n]{a - b} \neq \sqrt[n]{a} - \sqrt[n]{b}$$

RADICAL OF A PRODUCT

$$\sqrt{4 \cdot 9} \quad = \sqrt{4} \cdot \sqrt{9}$$

$$\sqrt[n]{a \cdot b} = \sqrt[n]{a} \cdot \sqrt[n]{b}$$

The radical of the product is the product of the radicals

$$\sqrt{4 + 9} \neq \sqrt{4} + \sqrt{9}$$

RADICAL OF A QUOTIENT

$$\sqrt[3]{\frac{64}{8}} = \frac{\sqrt[3]{64}}{\sqrt[3]{8}}$$

$$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

The radical of the quotient is the quotient of the radicals

FRACTIONAL EXPONENTS

$$27^{\frac{2}{3}} = \left(\sqrt[3]{27}\right)^2$$

$$a^{m/n} = \left(\sqrt[n]{a}\right)^m$$

Any number raised to a fractional exponent is equal to the **root** indicated by the **denominator** of the base raised to the power of the numerator

Compute $(25)^{-3/2}$

Simplify the expression

$$\left(\frac{25x^4y^{-5}}{x^{-6}y^3}\right)^{3/2}$$

Rationalize the denominator

$$\frac{3x}{\sqrt{x}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

SEATWORK